

FUNCTIONAL GRADIENTS IN THE INSULA DURING A MULTISENSORY INTEGRATION TASK

Lab Immersion 1

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1 Introduction

The insula is a highly researched cortical region that is both structurally and functionally heterogeneous and has extensive connections with cortical as well as subcortical structures. Known for its roles in salience detection, cognitive control, sensory processing and integration as well as emotional regulation and interoception, the insula is a key candidate of interest for multisensory integration tasks. A multisensory integration task containing visuotactile illusions is performed in virtual reality as the fMR images are collected using a corticospinal sequence. The current project aims to understand the role of the insula in driving visuotactile illusions or performing multisensory integration through the technique of functional gradients. Specifically, the group-averaged functional connectivity of the insula with itself and with the somatomotor cortex are used to compute gradient maps for resting state and different conditions of task. The observed functional hierarchy of the insula is then analyzed through the lens of specific hypothesis to understand how various insular sub-regions reorganize themselves during different levels of multisensory illusion.

2 Methods

2.1 Functional Connectivity in resting state and task

The timecourses of BOLD activity in the resting state (for the scan prior to task session) are extracted for whole-brain (Schaefer400), insular (custom atlas from Julich-brain cytoarchitectonic maps) and spinal cord (custom atlas from spinal cord toolbox maps of GM RoIs and WM tracts[1]) parcellations. The functional connectivity matrices for insula-insula connectivity and insula-SMC connectivity are computed using correlation measure for each of the 19 subjects and then averaged to get group functional connectivity matrices. Specifically for insula-SMC connectivity, first the insular timecourses are concatenated with the timecourse in the somatomotor RoIs defined by the Yeo 7-Network parcellation[2]. The Insula-SMC sub-part of the correlation matrix computed from the concatenated timecourse is taken as the Insula-SMC FC matrix.

For task condition, first the timings file was used to extract time ranges corresponding to the start and end of each individual block and these time ranges are converted to volumes for extracting block-wise timecourses for each parcellation. Similar steps as for rest are carried out to get the group-averaged functional connectivity matrices per task block.

A third type of FC matrix consisting of insula, spinal cord RoIs and RoIs of select cortical networks (namely Visual, Somatomotor and Limbic) was also constructed from a concatenated timecourse of activity in each of the constituent RoIs across the listed regions.

All analyses involving Nifti-image handling including extraction of timecourse for specific parcellations, were carried out using the Nilearn package in Python[3].

2.2 Gradients computation

The functional gradients are computed using the Brainspace toolbox in Python[4]. The pipeline includes a sparsification step to retain only top 10% columns per row of the FC matrix, application of cosine similarity kernel (cosine similarity of pairs of rows), and non-linear dimensionality reduction using diffusion map embedding ($\alpha = 0.5$) to obtain 10 components representing the functional gradients that can be projected back to insular RoI space.

For comparison between the gradients computed independently, Procrustes alignment is applied by aligning different gradient maps to a single reference axis(es). For Procrustes alignment, the `scipy.spatial.procrustes` function is used as it returns a disparity metric signifying level of mismatch between optimally aligned gradient map and the reference axes passed.

2.3 Mapping of insular RoIs to cyto-architectural sub-divisions

RoIs are mapped to specific cyto-architectural sub-divisions according to the definitions in [5] shown in ???. The sub-regions and their constituent RoIs are listed below:

- granular: Ig1,Ig2,Ig3,Id2
- agranular: Id1, Id3, Id5, Ia, Ia2, Ia3, Id8, Id9, Id10
- dysgranular: Id7, Id6, Id4

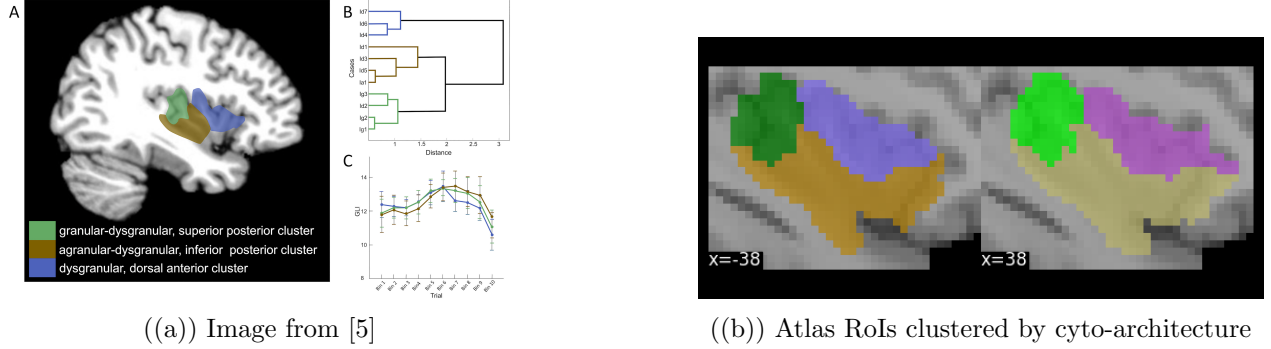


Figure 1: Grouping of insular RoIs to sub-regions defined by similarity of cytoarchitecture

2.4 Neurosynth decoding of clusters in gradient space

The RoIs distributed in 2D gradient space (top 2 gradients) are clustered using 3-means clustering and each of the clusters are used to create an activation map on which Neurosynth decoding is performed to obtain the top terms represented by the map in the Neuroquery dataset[6].

3 Results

3.1 Inter-hemispheric asymmetry in task gradient compared to rest

The first hypothesis relates to the asymmetry between left and right hemispheres in task condition compared to rest, arising from the lateralized visual and tactile stroking (of right index finger). To test this hypothesis, the principal gradient of both intrinsic insular gradients (gradients of insula-insula FC) as well as insula-SMC gradients were aligned to a left-to-right axis separating left hemisphere RoIs and the right hemisphere RoIs on either extreme (fig. 2).

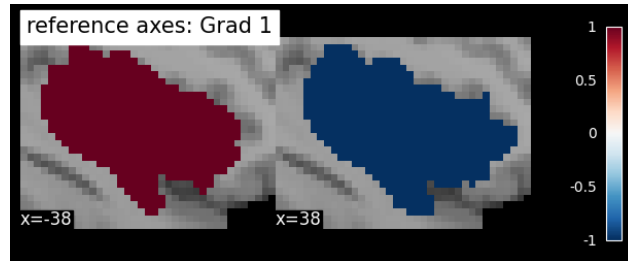


Figure 2: Reference Axis between left and right hemisphere RoIs

The result of aligning the gradient map at resting state is depicted below (fig 3). The maps for task conditions can be found in the appendix. The principal gradients at rest aren't asymmetric whereas

the ones in task are more asymmetric overall, as indicated by the lower measures of disparity with the reference.

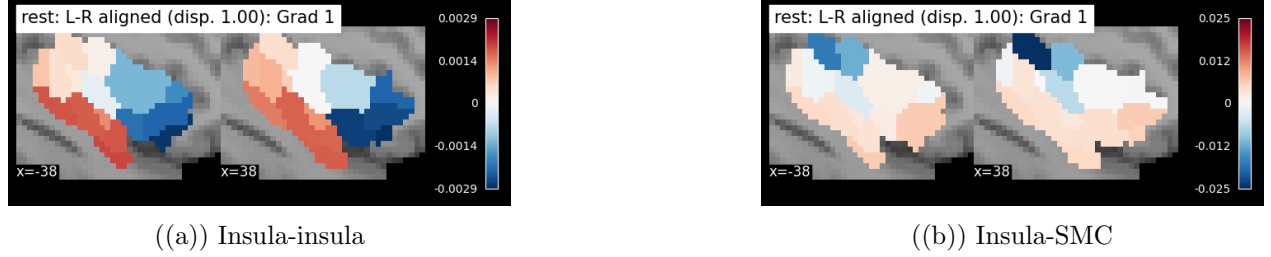


Figure 3: Principal gradient at rest aligned to L-R reference

The trend in disparity across trials are shown in figure 4. The intrinsic insular gradients are more asymmetric in embodiment than illusion conditions, and the symmetry is highest for mid and high illusion conditions specifically those following the two blocks containing low illusion (fig. 5). A closer look at the rank-shift plots of the trial-averaged maps with rank-shift plotted per cyto-architectural sub-division (fig 6(a)) reveals that the rank shift (distance from assigned ‘ideal’ rank on gradient hierarchy) is highest for dysgranular RoIs in High Illusion. i.e. left hemisphere dysgranular RoIs are closer to the right hemisphere RoIs in the functional hierarchy for high illusion - suggesting that there might be some bilateral recruitment of insular RoIs involved in integration of somatosensory input for higher order processing.

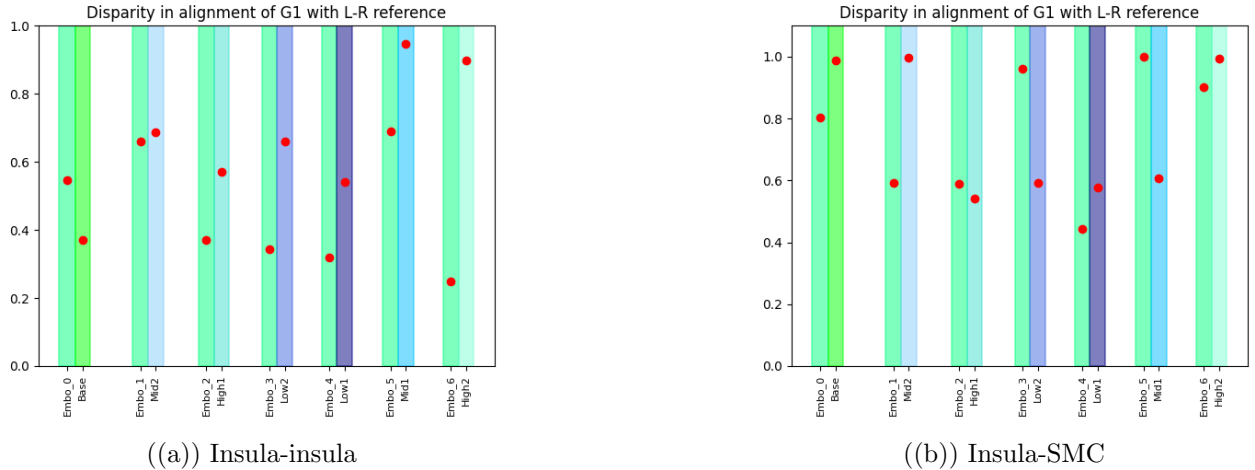
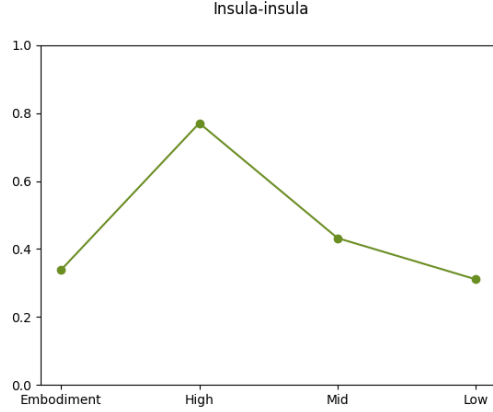
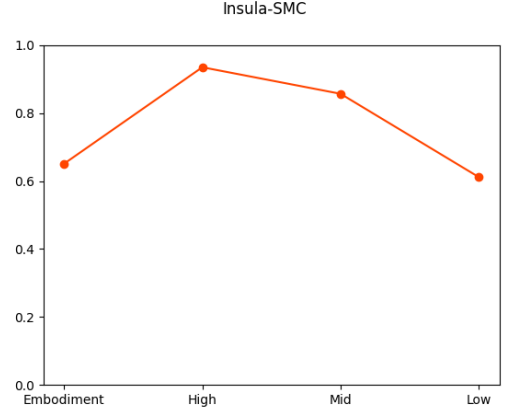


Figure 4: Disparity of alignment to L/R reference

Further, the insula-SMC gradients show a similar trend as insula-insula gradients on the trial-average case but to a much weaker extent. One potential explanation is that the insula-SMC FC might have a stronger tendency to isolate posterior insular regions known to be structurally connected to the SMC instead of emphasizing an inter-hemispheric difference. This could also lead to selective asymmetry only in certain insular RoIs (specifically posterior insula) with respect to functional connectivity with SMC while other insular RoIs do not get lateralized. From the rank-shift plot (fig. 6(b)), it seems that the rank shift for agranular RoIs are consistently high compared to the insula-insula case, suggesting that they are more symmetric in their functional relation to SMC. The same trend of left dysgranular RoIs shifting towards right hemisphere that is seen in intrinsic insular gradients, persists in the insula-SMC case as well.

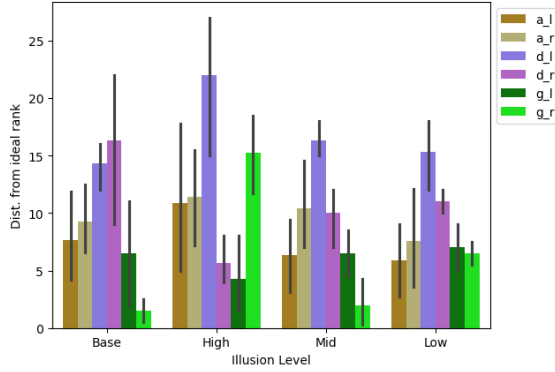


((a)) Insula-insula

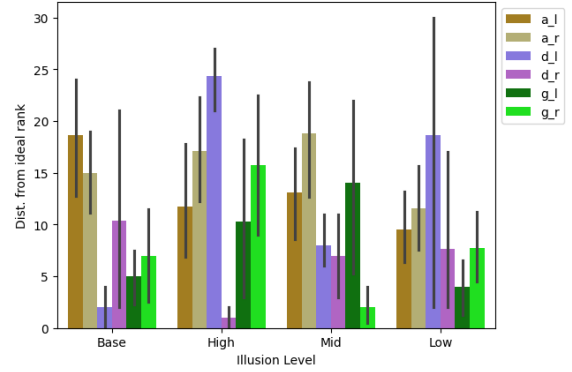


((b)) Insula-SMC

Figure 5: Disparity of alignment to L/R reference - Trial-averaged



((a)) Insula-insula



((b)) Insula-SMC

Figure 6: Rank-shift of each cyto-architectural sub-division in trial-averaged gradients. Rank shift for a given RoI is computed as $|rank_{G1} - rank_{L/R}|$ where $rank_{G1}$ is the rank of the RoI on the functional axis after alignment and $rank_{L/R}$ is 0 if the RoI is in the left hemisphere and 31 otherwise.

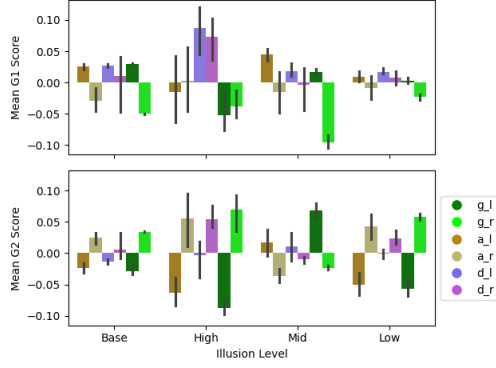
3.2 Cluster-wise average gradient scores

Figure 7 depicts the gradient scores per cyto-architectural sub-division when the top 2 gradients are aligned to resting state. In fig. 7(a), the gradient 2 shows a Left-Right axis (most prominently for high illusion). The gradient 1 of high illusion appears to distinguish dysgranular and granular cytoarchitectures. However, when we look at the insula-SMC gradients (fig. 7(b)) there isn't a similar clear trend visible.

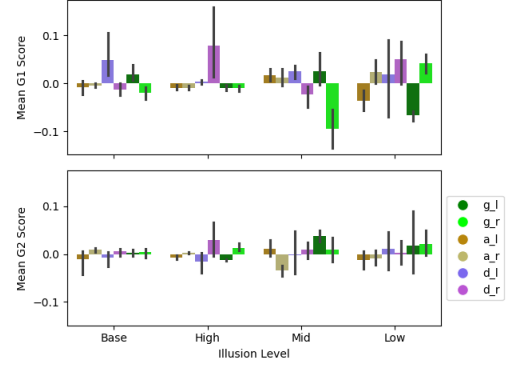
3.3 Clusters in task gradient space are functionally heterogeneous

The clusters projected to insular space and their corresponding neurosynth decoding terms are shown in figures 8 and 9. The clustering of points in gradient space are depicted in figure ?? in the appendix.

In resting state, the clusters are highly symmetric across hemispheres as expected intuitively from the symmetry in gradient maps. Furthermore, the individual clusters distinguish the dorsal posterior, dorsal anterior and ventral insula. The neurosynth terms are suggestive of the functional homogeneity within each cluster - the dorsal anterior includes terms related to integration whereas



((a)) Insula-insula



((b)) Insula-SMC

Figure 7: Average gradient score per cyto-architectural sub-division in insular gradients. In each case, the gradients are aligned to respective top 2 gradients in resting state

the dorsal posterior includes terms linked to somatosensation and pain and the ventral cluster includes terms related to emotional valence and processing.



Figure 8: Clusters obtained using 3-Means in 2D gradient space of resting insula-insula FC. Individual clusters are converted to cluster activation maps on which Neurosynth decoding is performed

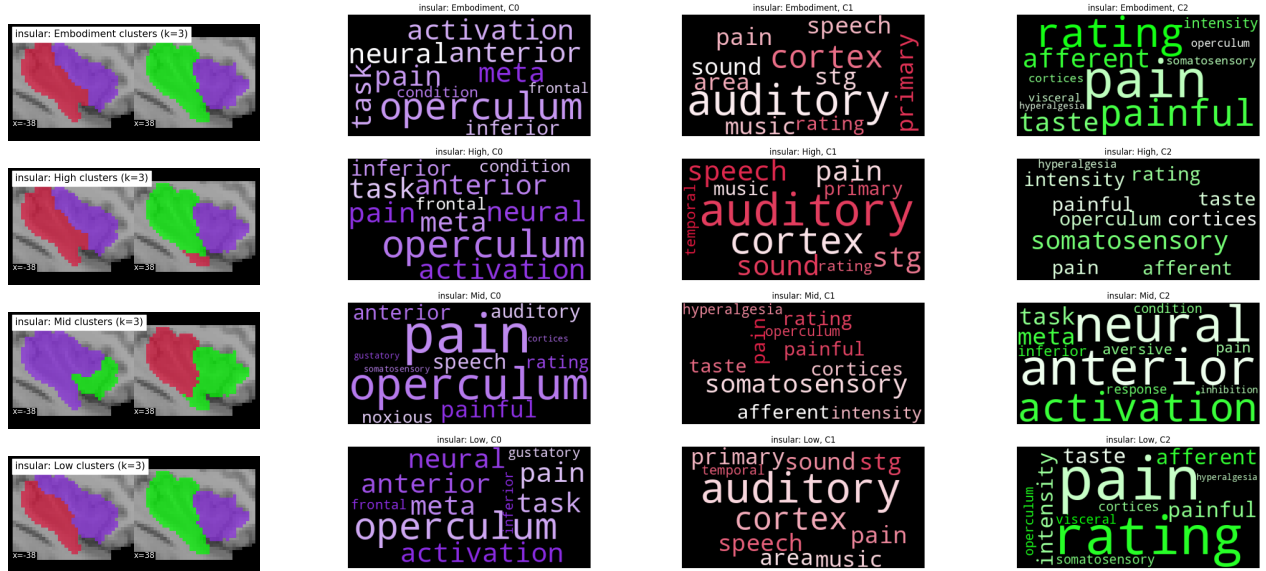


Figure 9: Clusters and neurosynth decoding for intrinsic insular gradients of task FC

3.4 Cortical and spinal RoIs are least functionally segregated in low illusion

To understand how cortical and spinal RoIs are organized between each other in the functional hierarchy during rest and various task conditions, the gradient method is applied to a functional connectivity matrix consisting of selected cortical networks (Limbic, Visual, Somatomotor), the insula, and spinal cord RoIs (including white matter tracts and gray matter RoIs from levels C1 to T1).

The distribution plots in fig. 10 show that the cortical and spinal RoIs are well separated in both task as well as resting state. However, in low illusion, the spinal and cortical RoIs appear less segregated and the spinal RoIs occupy a larger extent on the shared hierarchy as compared to other conditions.

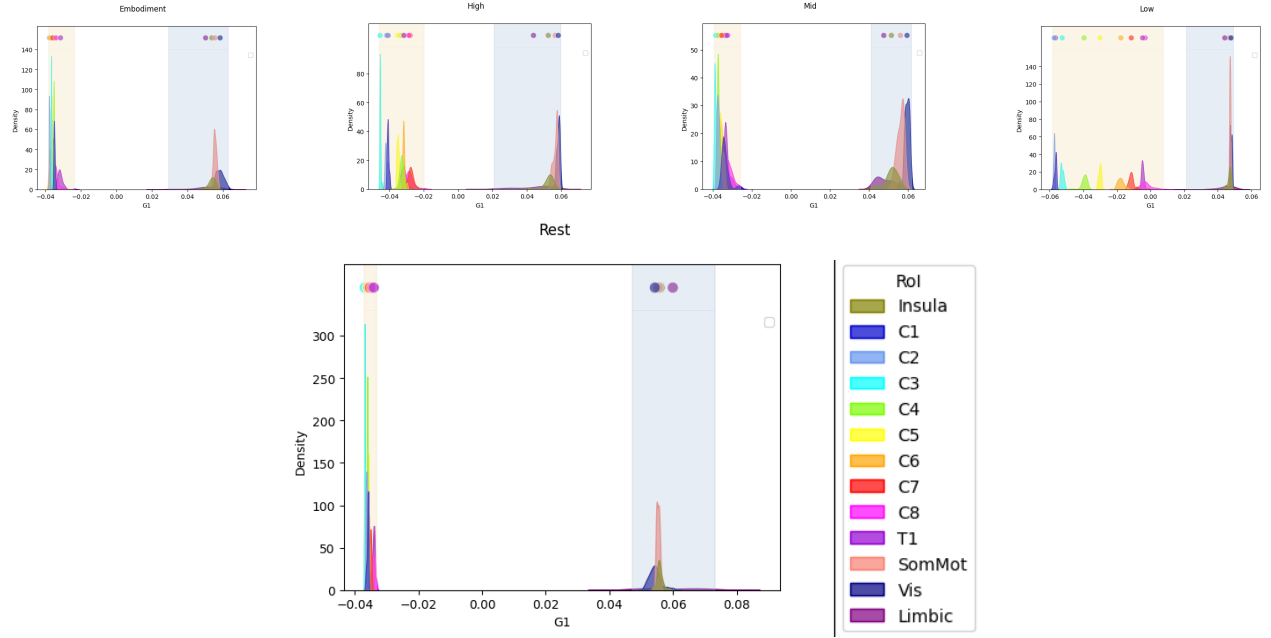


Figure 10: Distribution of RoIs along principal gradient. The yellow shaded region represents extend of spinal RoIs and blue represents cortical RoIs. All task gradients are aligned to principal gradient in resting state. The scatter points on the top of the plots indicate the location of RoIs' average location in 1D gradient space.

4 Discussion

This project aimed to explore how the functional hierarchy of the insula reorganizes during the course of a multisensory integration task. To this end, the functional gradients of insular functional connectivity with itself as well as with the somatomotor network were computed in both rest as well as different conditions of said task.

First, the gradients were analyzed for their left-right asymmetry by aligning the principal gradient to a left-right axis which enabled quantification of the degree of asymmetry by means of a disparity score. The intrinsic insular gradients, i.e. the gradients of insula-insula FC demonstrated left-right symmetry in rest and higher asymmetry in task indicated by higher and lower disparity scores respectively. Among different task conditions, the high and mid illusion conditions displayed lower left-right symmetry than embodiment and low illusion in the gradients of trial-averaged FC. On the other hand, gradients of insula-SMC FC displayed lower asymmetry overall while maintaining

a similar trend of higher asymmetry in task and lower asymmetry for high and mid illusion compared to other task conditions. To explore the reason for these trends namely lower asymmetry in high illusion and lower asymmetry for insula-SMC gradients, the rank shift was computed for each RoI and plot according to the cyto-architectural sub-division they belonged to. In intrinsic insular gradients, highest rank shift was observed for dysgranular RoIs of the left hemisphere in the high illusion condition. This is suggestive of a reorganization of intrinsic insular functional hierarchy in high illusion where the dysgranular RoIs of both hemispheres group together on one end while the granular on the other, as corroborated by the trend of averaged gradient score in fig. 7(a). This likely suggests a dysgranular-granular functional axis emerges in high illusion where there is likely high 'internal model error' when integrating sensory inputs with sensory prediction. The insula-SMC gradients show a higher rank shift for all agranular RoIs in all conditions - suggesting the insular functional connectivity with SMC may be agnostic of the agranular RoIs with which there is no direct structural connectivity. The dysgranular RoIs of the left hemisphere still display higher rank shift in high illusion suggesting a similar hypothesis of functional reorganization as in the case of intrinsic insular gradients.

Second, the clustering of insular RoIs in 2D gradient space (for insula-insula FC) by 3-means demonstrated functional heterogeneity of individual gradient space clusters as opposed to functionally distinct clusters in resting state, with within cluster functional heterogeneity indicated by neurosynth terms from diverse cognitive processes as opposed to related terms involved in a single process.

Finally, to investigate reorganization of the collective corticospinal hierarchy, the functional connectivity including selected resting-state networks (namely visual, somatomotor and limbic), the insula and spinal cord RoIs is computed and the principal gradient is analyzed. Qualitative plots indicate a clear functional segregation of cortical and spinal RoIs for both resting and task conditions while the degree of segregation varies between the two. The segregation is lower in task compared to rest and the lowest segregation is for the low illusion condition. Further, the spinal part of the hierarchy occupies most extent along gradient hierarchy in low illusion as compared to other conditions. The individual spinal levels' gradient score distribution also appear more segregated in low illusion compared to other conditions. This indicates more 'cross-talk' between spinal and cortical systems in low illusion where there is low sensory ambiguity.

There are several limitations to this project. Firstly, all gradient analyses and subsequent inferences are based on qualitative inspections of the functional gradient computed from group-averaged functional connectivity. In order to test these inferences, data-driven approaches such as iCAPS or bootstrapping to establish significance of the present results are required. For the functional hierarchy including cortical and spinal RoIs, the functional segregation between cortical and spinal parts need to be quantified for each task condition to understand if the difference in segregation between conditions is actually significant. Furthermore, looking into individual-level gradients and their correlation with the subjective perception of illusion throughout the task may also reveal interesting findings. Secondly, the clustering and neurosynth decoding of cluster activation maps was done using 3-Means clustering for each gradient separately without any matching between the clusters. A preferred approach would be to match the task gradients all to resting state clusters. Moreover, the Neurosynth terms have to be filtered to remove redundant terms to show more relevant terms. Thirdly, the analyses based on cyto-architectural sub-divisions of the insula assumed that the 32 RoIs can be distributed between sub-divisions as fig. 1(b), which requires validation and matching with more than one source from literature. Further, the findings are biased by the fact

that the current sub-divisions include a much higher number of ‘agranular’ RoIs than ‘granular’ or ‘dysgranular’. An alternate approach would be to explore other insular sub-divisions such as known well-established functional sub-divisions of the insula.

5 Future Steps

Investigating the movement of different insular RoIs to reorganize functional hierarchy throughout the task flow using qualitative and quantitative approaches could reveal more insights. It could be useful to approach this through a more specific hypothesis for underlying cognitive processes being manipulated by the insula - multisensory integration and the role of insula as a salience detection node throughout the task, or insula as a driver of illusion in high sensory ambiguity to allow consistency with an internal model.

6 Conclusion

This project explored whether functional gradients in the insula can reveal the reorganization of its functional hierarchy in different conditions of a multisensory integration task involving simultaneous visual and tactile stimulation of varying levels of congruence. The intrinsic insular gradients were found to be asymmetric in task setting compared to rest, with varying levels of asymmetry for different levels of illusion within the task. Insular RoIs were grouped into sub-regions based on similarities in cytoarchitecture to investigate whether the reorganization is guided by this similarity or disrupts the existing grouping within the insula. The hierarchy of corticospinal RoIs in task vs. rest revealed different degrees of functional segregation and relative extent between the cortical and spinal parts of the combined axis. Further validation steps are necessary to establish the inferences made from the current results.

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